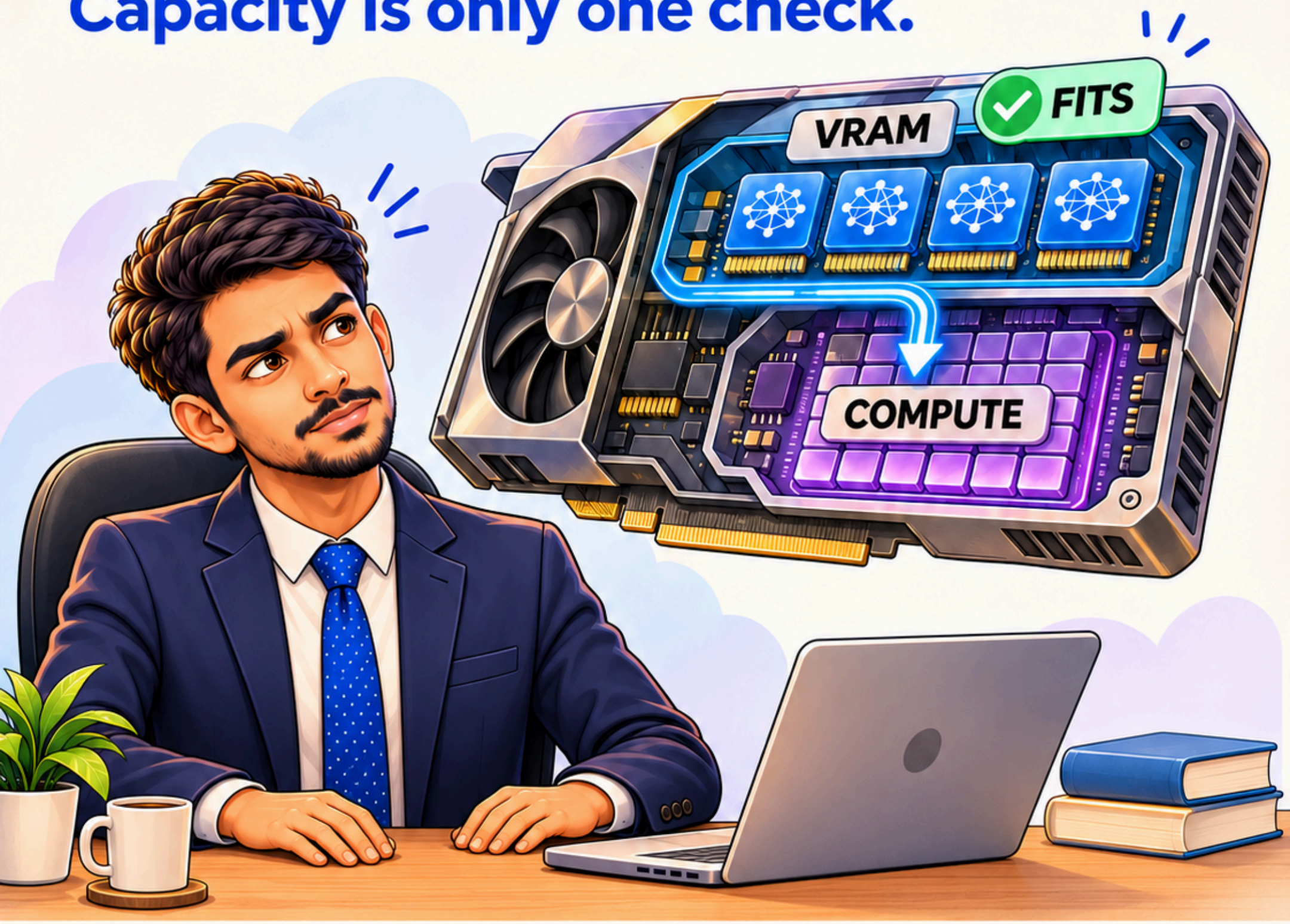
AI SYSTEMS

The model fits. Why is it still slow?

Capacity is only one check.



Capacity. Bandwidth. Compute.

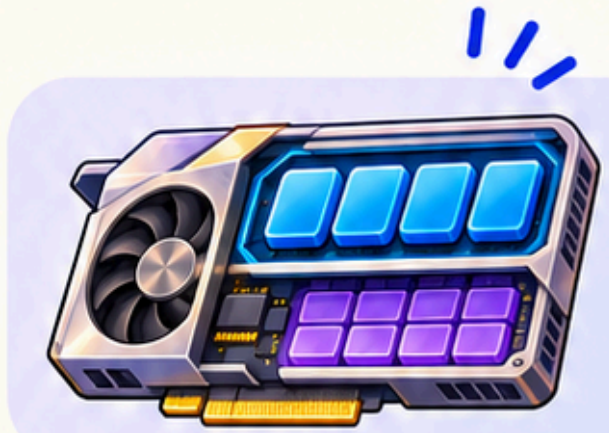
AI SYSTEMS

What does ‘the model fits’ mean?

Model weights are the **learned numbers** the model uses to make predictions.

Loading the model puts these numbers into **VRAM**: the GPU’s main memory.





The workload also needs room for saved context and calculation buffers. If all of that fits, you have passed a storage check.



Capacity is measured in **GB (gigabytes)**.

Storage capacity answers: is there enough room?

AI SYSTEMS

What happens for one token?



A token is a chunk of text, not always a whole word.



1. READ

GPU units read weights from VRAM in pieces, through small on-chip stores called caches and registers.



2. CALCULATE

GPU compute units use those numbers to calculate the next-token probabilities.



3. APPEND

The system selects a token and adds it to the response.



Memory bandwidth is the amount of data moved per second, measured in GB/s.

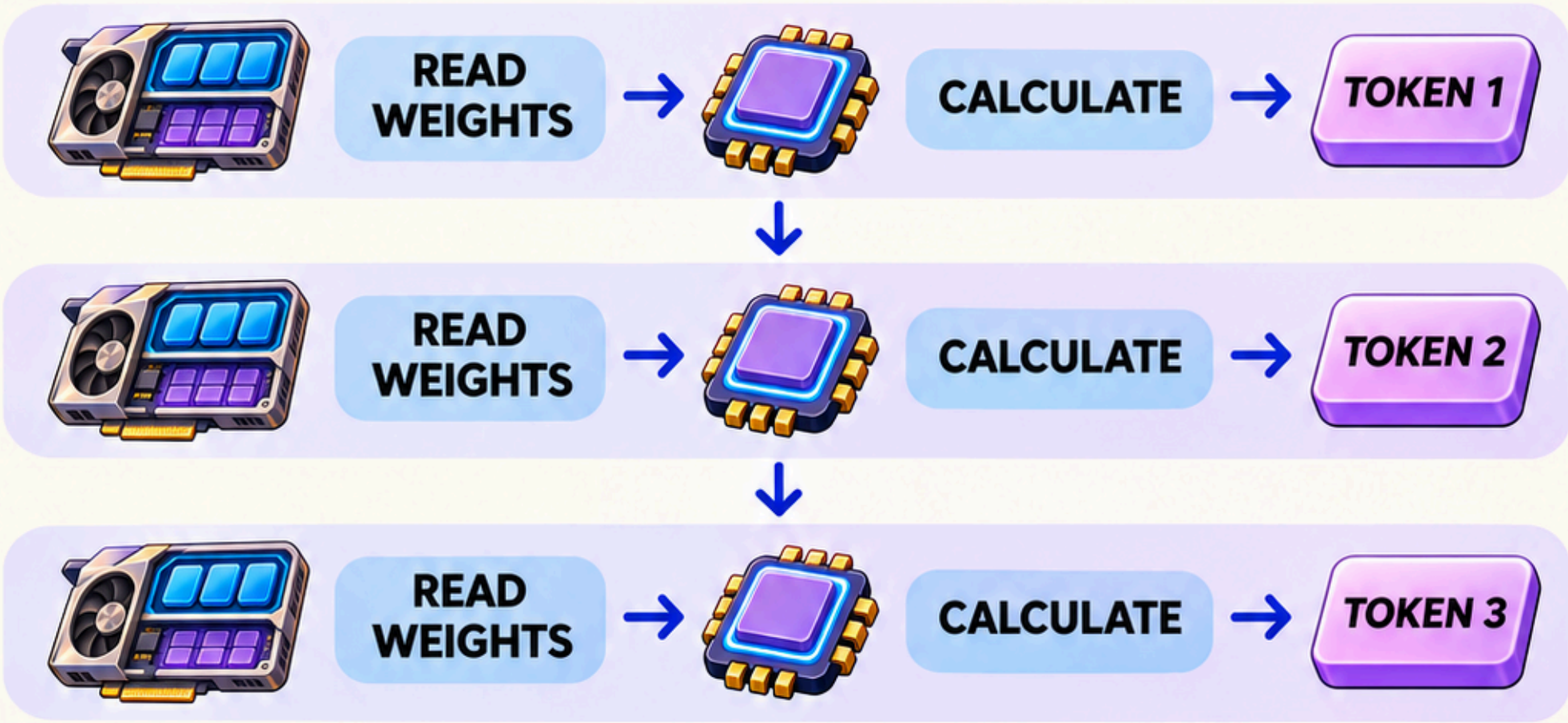
The model must move data and perform math.

AI SYSTEMS

Now repeat for the next token.



Ordinary generation produces **one token at a time.** Each next token uses the required weights again.





A large model cannot stay entirely in the GPU's **small on-chip stores.** The required weights are read again from VRAM.



For **one response at a time,** moving these weights can take longer than the calculations.

Then memory bandwidth limits token speed.

AI SYSTEMS

Same capacity. Different read times.



Suppose each GPU has **24 GB** of VRAM. Both workloads fit. Each step reads **14 GB** of weights.

EXAMPLE A	EXAMPLE B
MEMORY BANDWIDTH	MEMORY BANDWIDTH
350 GB/s	**700 GB/s**
14 GB ÷ 350 GB/s	14 GB ÷ 700 GB/s
= 0.04 s = 40 ms	**= 0.02 s = 20 ms**



ms means milliseconds.



Ideal weight-read times only. This is illustrative arithmetic, not a benchmark. Other work also affects token latency.

Enough space does not guarantee fast data access.